A Synthetic Data and Random Forest Framework

for Psychiatric Risk Stratification in Multi-Modal Monitoring Systems

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Pre-Publication Notice

This document is a **pre-publication draft (Version 0.1)** prepared for academic review and interdisciplinary collaboration.
The author holds a Ph.D. in City and Regional Planning with a research focus on mathematical modeling, complex adaptive systems, and urban informatics. The framework presented here lies at the intersection of spatial cognition, behavioral simulation, and machine learning, offering a prototype for simulating and evaluating psychiatric risk profiles in synthetic populations.
Although the study integrates constructs from psychiatry—such as mood dynamics, emotional congruence, and psychometric evaluation—it is intended solely as a computational artifact. The system does not provide diagnostic recommendations, therapeutic guidance, or clinical interpretations, and should not be used for patient care or mental health treatment.
The simulated patients, behavioral data, and risk profiles used in this framework are entirely artificial. They are designed to support methodological innovation in cognitive informatics, ethical AI development, and scalable experimentation for digitally augmented behavioral health systems.
Readers are encouraged to interpret the findings through the lens of interdisciplinary research, blending urban systems thinking with computational psychiatry to explore novel directions in digital public health and well-being analytics.

Abstract

This paper introduces a computational pipeline for generating synthetic psychiatric patient data and classifying their clinical risk using a Random Forest model. The system simulates affective, behavioral, and psychometric indicators across time, enabling risk-grade labeling from I (low risk) to V (critical risk). A supervised learning model is trained on these features to predict risk levels, while interpretability is ensured using SHAP analysis. The proposed architecture enables scalable experimentation for digital phenotyping

and early intervention strategies in mental health.

1 Introduction

1.1 The Data Bottleneck in Computational Psychiatry

The application of machine learning (ML) in psychiatry holds transformative potential, particularly for early risk detection and personalized intervention. However, this potential is persistently limited by the sensitive and scarce nature of psychiatric datasets. Real-world mental health data is difficult to acquire due to ethical constraints, stringent privacy regulations (e.g., HIPAA, GDPR), stigma-related underreporting, and variability in symptom presentation. Moreover, existing datasets are often small, temporally sparse, and institution-specific—resulting in limited generalizability and model instability. These factors collectively contribute to a reproducibility bottleneck in computational psychiatry.

1.2 The Promise of Synthetic Data

Synthetic data generation offers a compelling solution to this bottleneck. By simulating patient-like trajectories using rule-based and stochastic methods, researchers can produce structurally valid and ethically safe datasets. These synthetic profiles can model dynamic behaviors and multimodal signals—including affective states, psychometric patterns, activity rhythms, and natural language features—without exposing identifiable personal data. This enables scalable experimentation, hypothesis testing, and benchmark creation in a domain traditionally limited by access and risk sensitivity.

1.3 Study Objectives

This study introduces a generative and predictive pipeline for psychiatric risk modeling using synthetic data. The objectives are as follows:

- To develop a generative framework that produces realistic, label-controllable psychiatric profiles incorporating multimodal behavioral and affective signals.
- To train a Random Forest classifier on these synthetic profiles to predict risk stratification levels from Grade I (low-risk) to Grade V (critical-risk).
- To apply SHAP (SHapley Additive exPlanations) methods to interpret feature importance and enhance model transparency.

1.4 Contributions

The framework provides a reproducible infrastructure for simulating patient diversity, modeling clinical heterogeneity, and evaluating ML performance under controlled data generation scenarios. By aligning cognitive informatics, behavioral modeling, and explainable machine learning, the proposed system supports the development of ethically compliant, scalable tools for digital mental health research and prototyping.

2 Related Work

2.1 Synthetic Data Generation in Psychiatry

Synthetic data generation has emerged as a practical alternative to real-world clinical datasets, especially in domains where privacy, consent, and reproducibility pose significant constraints. In psychiatry and clinical psychology, synthetic data approaches are used to simulate mood trajectories, simulate psychometric test outcomes, and emulate daily behaviors such as sleep, activity, and social interaction.

Two main paradigms dominate this space: rule-based simulation and deep generative modeling. Rule-based systems rely on expert-defined probability distributions, temporal patterns, and feature correlations to simulate plausible patients. For example, a high-anxiety synthetic profile might be configured with elevated mood variance, disrupted circadian rhythm, and elevated neuroticism scores. Deep learning approaches, including Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), have also been explored for realistic patient state synthesis. However, these models often lack interpretability and require large training datasets, which are limited in psychiatric domains.

Our framework adopts a modular rule-based strategy to enable feature-level control over synthetic profiles. This allows precise manipulation of risk gradients, emotional volatility, and psychometric variability in ways that mirror observed clinical patterns while preserving mathematical tractability and transparency.

2.2 Machine Learning for Digital Mental Health

Machine learning models—particularly ensemble methods—have been widely used in digital mental health studies to classify emotional states, detect risk, and predict psychiatric outcomes. Random Forests (RF), as an ensemble of decision trees trained on bootstrapped feature sets, offer several advantages: high accuracy, resistance to overfitting, robustness to noise, and native feature importance estimation.

RF models operate by computing a majority vote (classification) or mean prediction (regression) across independently trained decision trees. Let $f_i(\mathbf{x})$ denote the prediction of the i-th tree on input feature vector $\mathbf{x}$. Then, the RF prediction is:

$$\hat{y}_{RF}(\mathbf{x}) = \frac{1}{T} \sum_{i=1}^{T} f_i(\mathbf{x})$$

where T is the total number of trees. For risk stratification, each input $\mathbf{x}$ corresponds to a synthetic patient profile consisting of affective, cognitive, behavioral, and physiological dimensions.

Interpretability of RF models is often enhanced using SHAP (SHapley Additive exPlanations), which provide per-instance attributions by decomposing the model output as a sum of feature contributions:

$$\hat{f}(\mathbf{x}) = \phi_0 + \sum_{j=1}^{n} \phi_j$$

where ϕ_j is the contribution of feature j and ϕ_0 is the expected model output. This decomposition is grounded in cooperative game theory and provides a principled approach to interpret model predictions for individual synthetic patients.

2.3 Psychiatric Risk Grading and Composite Metrics

Psychiatric risk stratification commonly relies on composite assessments derived from clinical interviews, standardized psychometric instruments (e.g., PHQ-9, GAD-7, HAM-D), and observational data. Risk is rarely a function of a single dimension; rather, it emerges from interrelated affective, behavioral, and cognitive indicators.

In clinical settings, stratification often follows heuristic rules or weighted scoring systems. For example, suicidality risk may be classified into five levels (I–V) based on severity, frequency, and co-morbid symptoms. Similarly, DSM-5-based assessments involve functional impairment scaling across social, occupational, and interpersonal domains.

To mimic this structure, we define a scalar composite danger score $D \in [0, 1]$ using a weighted function of normalized features:

$$D = \sigma\left(\sum_{k=1}^{K} w_k \cdot \tilde{x}_k\right)$$

Here, $\tilde{x}_k$ are normalized features (e.g., mood variance, sentiment polarity, sleep quality), $w_k \in \mathbb{R}$ are risk weights, and $\sigma(\cdot)$ is a sigmoid function that maps raw risk to a probability-like output. The continuous danger score is discretized into grades (I–V) using quantile thresholds, enabling classification-compatible targets.

This approach aligns with clinical reasoning while remaining computationally viable, allowing synthetic risk labels to capture both complexity and interpretability.

3 Synthetic Data Generation Framework

3.1 Patient Identity Simulation

Each synthetic patient P_i is modeled as a time-indexed multivariate profile over T days, defined by:

$$P_i = \{M_t, S_t, L_t, E_t, F_t\}_{t=1}^{T}$$

where:

- $M_t \in \mathbb{R}^5$: 5-dimensional mood vector (e.g., anxiety, depression, calmness, irritability, motivation)
- $S_t \in \mathbb{R}^2$: sentiment polarity and subjectivity score extracted from simulated daily journal or speech
- $L_t \in \mathbb{R}^2$: physiological metrics such as sleep duration (hours) and step count
- $E_t \in \mathbb{R}^n$: emotional label distribution (from simulated video/text/audio input), such as Ekman or Plutchik classes
- $F_t \in \mathbb{Z}^{d_f}$: responses to psychometric forms (e.g., PHQ-9, GAD-7)

Each vector evolves stochastically or deterministically over time to reflect plausible human affective-behavioral dynamics, modulated by risk level $G \in \{I, II, III, IV, V\}$.

3.2 Affective Dynamics via Sinusoidal Perturbation

Daily mood components are generated using a sinusoidal perturbation model:

$$m_{k,t} = \text{clip}\left(B_k + A_k \cdot \sin\left(\frac{2\pi t}{P}\right) + \epsilon_t,\ 1,\ 5\right)$$

where:

- $B_k \in [1, 5]$: baseline mood intensity for mood type k
- A_k: amplitude of emotional fluctuation
- $P \in \mathbb{Z}^+$: periodicity (e.g., circadian = 1, biphasic = 7, cyclic = 14)
- $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$: Gaussian noise capturing stochastic variance

The `clip` function ensures mood values remain within bounded clinical scales (1–5 Likert scale). Different psychiatric archetypes (e.g., bipolar, depressive, anxious) are parameterized with distinct B_k, A_k, and P values.

3.3 Risk-Conditioned Sampling

To reflect diagnostic heterogeneity, patient features are sampled conditional on risk level $G \in \{I, II, III, IV, V\}$, where:

- Mood score averages $\mu_{M,G} \in [1, 5]$ are shifted downward with increasing G
- Sentiment polarity $\mu_{S,G} \in [-1, 1]$ becomes more negative
- Sleep-step vectors $L_t \sim \mathcal{N}(\mu_{L,G}, \Sigma_L)$ reflect dysregulation (e.g., insomnia, fatigue)
- Emotion label vectors $E_t \sim \text{Multinomial}(\boldsymbol{p}_G)$, where $\boldsymbol{p}_G$ defines risk-specific emotion probabilities (e.g., higher anger/sadness in $G = IV$)

This conditional logic allows generation of clinically plausible trajectories based on assumed risk state, enabling downstream supervised learning.

3.4 Psychometric Form Score Generation

Psychometric scores are generated weekly or monthly using a severity-modulated function:

$$f_{j,t} = \text{round}\left(b_j \cdot v_j \cdot \phi_t + \epsilon_j\right)$$

with:

- $b_j \in \mathbb{R}^+$: base severity for item j (e.g., PHQ-9 Q1: anhedonia)
- $v_j \sim \mathcal{U}[0.8, 1.3]$: patient-specific vulnerability trait for item j
- $\phi_t \in [0, 1]$: mood-linked modulation function (e.g., inverse of calmness or motivation)
- $\epsilon_j \sim \mathcal{N}(0, \sigma_j^2)$: psychometric response noise

Final scores are mapped into clinical severity bins (e.g., minimal, mild, moderate, severe) based on DSM-aligned cutoffs. These scores feed into the risk grading function described in Section 4.

4 Functional and Risk Scoring

4.1 Composite Functioning Score

A patient's ability to maintain stable behavioral routines is a core indicator of mental well-being. To quantify this in a clinically meaningful way, we define a composite **Functioning Score** $F_t \in [0, 100]$, calculated daily for each synthetic patient P_i at time t. This score penalizes observable behavioral dysfunction across multiple modalities, each with a clinically informed threshold θ_i. Formally:

$$F_t = 100 - \sum_{i=1}^{K} w_i \cdot \mathbb{I}(x_i < \theta_i)$$

where:

- x_i: the value of a behavioral or emotional feature (e.g., number of steps, mood, sleep duration)
- θ_i: feature-specific health threshold (e.g., 3000 steps/day)
- w_i: clinically motivated weight (penalty severity for crossing threshold)
- $\mathbb{I}(\cdot)$: indicator function returning 1 if condition is met, 0 otherwise

This rule-based design imposes an interpretable and explainable penalty system: each deviation from a health norm subtracts a weight from the maximum functioning score of 100.

The following six dimensions are evaluated per day:

1. **Physical Activity:** Step count x_1, with $\theta_1 = 3000$ steps/day
2. **Sleep Duration:** Nightly hours x_2, with $\theta_2 = 6$ hours
3. **Mood Level:** Mean of 5D mood vector x_3, with $\theta_3 = 2.5$
4. **Sentiment Polarity:** NLP-based score x_4, with $\theta_4 = -0.2$
5. **Affective State:** Dominant categorical emotion from video/audio (e.g., sadness or anger)
6. **Form-Based Severity:** Psychometric scores exceeding thresholds (e.g., PHQ-9 $\geq$ 10, GAD-7 $\geq$ 8)

Each penalized condition contributes a decrement $w_i \in \{5, 10, 15\}$ to the overall score, based on clinical impact (e.g., high PHQ-9 is weighted more heavily than low sleep).

A score of $F_t = 100$ denotes optimal daily functioning. Lower scores indicate increased behavioral impairment, emotional dysregulation, or clinical risk—mirroring real-world psychiatric decline.

4.2 Danger Score and Risk Grade Assignment

While functioning captures behavioral stability, it must be combined with mood and clinical symptomatology to estimate risk. To this end, we define a scalar **Danger Score** $D_t \in [0, 1]$ that merges three critical streams:

$$D_t = \alpha \cdot \left(1 - \frac{\bar{m}_t}{5}\right) + \beta \cdot \left(1 - \frac{F_t}{100}\right) + \gamma \cdot S_f$$

where:

- $\bar{m}_t \in [1, 5]$: daily average mood (5D vector collapsed to mean)
- $F_t \in [0, 100]$: composite functioning score from Section 4.1
- $S_f \in [0, 1]$: normalized severity signal from forms (e.g., PHQ-9/GAD-7/PSS-10)
- α, β, γ: model coefficients controlling feature importance; typically $\alpha + \beta + \gamma = 1$

This formulation ensures:

- Low mood ($\bar{m}_t \downarrow$) increases D_t
- Poor functioning ($F_t \downarrow$) increases D_t
- High psychometric severity ($S_f \uparrow$) increases D_t

Weights α, β, γ can be tuned to model different clinical priorities—e.g., acute emotion detection vs. chronic behavioral patterns.

Grade Mapping from Danger Score

To interpret the continuous danger score categorically, we define risk grades based on fixed percentile cutoffs:

$$G_t = \begin{cases} \text{Grade I (Low)}, & D_t < 0.2 \\ \text{Grade II (Mild)}, & 0.2 \leq D_t < 0.4 \\ \text{Grade III (Moderate)}, & 0.4 \leq D_t < 0.6 \\ \text{Grade IV (High)}, & 0.6 \leq D_t < 0.8 \\ \text{Grade V (Critical)}, & D_t \geq 0.8 \end{cases}$$

These thresholds can be empirically tuned to match population-level risk prevalence, or applied heuristically for simulation purposes.

4.3 Visualizing the Danger Score Surface

To understand the behavior of the danger function, we visualize D_t over the joint domain $(\bar{m}_t, F_t) \in [1, 5] \times [0, 100]$. The severity penalty S_f is held constant at 0.5. The resulting surface illustrates how different mood-functioning combinations translate into risk.

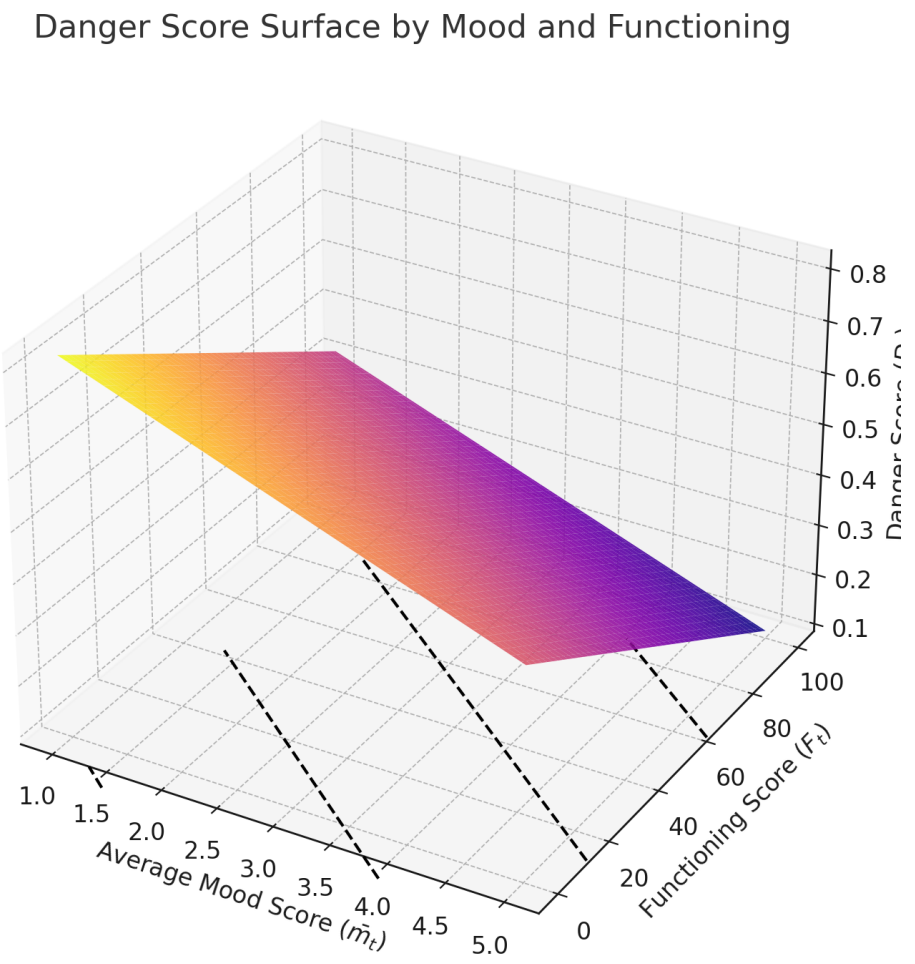


Figure 1: Danger score surface D_t as a function of average mood $\bar{m}_t$ and functioning score F_t, with a constant severity term $S_f = 0.5$. Dashed contour lines delineate grade thresholds (I–V).

The surface is non-linear and separable, making it suitable for interpretability and sampling. Low mood and low functioning interact synergistically to drive the risk above critical thresholds, consistent with clinical intuition.

5 Random Forest Classifier for Risk Prediction

5.1 Feature Vector Design

Each synthetic patient-day $x_i \in \mathbb{R}^d$ is transformed into a structured feature vector composed of both raw signals and engineered summary statistics. The design incorporates multimodal behavioral, emotional, and psychometric indicators as follows:

- **Mood Dynamics:**
  - Mean and standard deviation of 5D mood vector over past T days
  - Min/max values and daily slope trends (i.e., temporal volatility)
- **NLP-Based Features:**

  - Sentiment polarity $s_t \in [-1, 1]$
  - Subjectivity index $u_t \in [0, 1]$
  - Emotion vector logits (anger, sadness, joy, etc.)

- **Physiological and Behavioral:**
  - Step count (daily average and rolling 3-day trend)
  - Sleep hours (mean, deviation)
  - Sleep/step regularity (entropy or coefficient of variation)

- **Psychometric Severity:**
  - Raw form scores (PHQ-9, GAD-7, PSS10)
  - Binary threshold exceedance flags

- **Composite Scores:**
  - Functioning score $F_t \in [0, 100]$
  - Danger score $D_t \in [0, 1]$

- **Contextual Encodings:**
  - Day-of-week and weekend flag
  - Lagged grade from previous day (if sequential modeling is used)

The final input vector thus captures both short-term state and cumulative risk signals, designed to train robust classifiers on synthetic patients emulating real psychiatric profiles.

5.2 Model Training and Decision Structure

We train a supervised classifier to map feature vectors x_i to categorical risk labels $y_i \in \{\text{I}, \text{II}, \text{III}, \text{IV}, \text{V}\}$. For this task, we employ a **Random Forest (RF)** model due to its suitability for tabular, heterogeneous, and high-dimensional data.

The prediction is made by majority vote across B decision trees h_b:

$$\hat{y}_i = \text{mode}\left\{h_b(x_i)\right\}_{b=1}^{B}$$

Each decision tree h_b is constructed using bootstrap sampling of the training data and split based on **Gini impurity**:

$$Gini(S) = 1 - \sum_{c=1}^{C} p_c^2$$

where p_c is the proportion of class c among samples at node S. At each node, the algorithm chooses the feature and split value that minimizes the weighted impurity across child nodes.

Training Procedure:

- Data is split using *stratified sampling* to preserve grade distribution in both train and test sets.

- Synthetic samples from each risk grade G_t are balanced to mitigate class imbalance.
- RF is trained on 80% of samples and validated on 20%, optionally using cross-validation for hyperparameter search.

5.3 Hyperparameters and Model Tuning

The model's behavior is governed by several key hyperparameters:

- **Number of trees** $B \in \{100, 200, 500\}$: More trees reduce variance at cost of computation.
- **Maximum depth** $D \in \{10, 20, None\}$: Controls overfitting.
- **Minimum samples per split**: Prevents small node over-specialization.
- **Feature bagging ratio**: Determines how many features are randomly sampled at each node.
- **Class weight tuning**: Optional reweighting if grades are skewed.

Random Forests are non-parametric, robust to outliers, and suitable for interpretation via feature importance metrics (discussed in Section 6).

5.4 Visualization of Classifier Behavior

To gain insight into the behavior and boundaries learned by the Random Forest model trained on synthetic psychiatric data, we provide three visualization panels that collectively capture clustering, classification confidence, and model complexity effects.

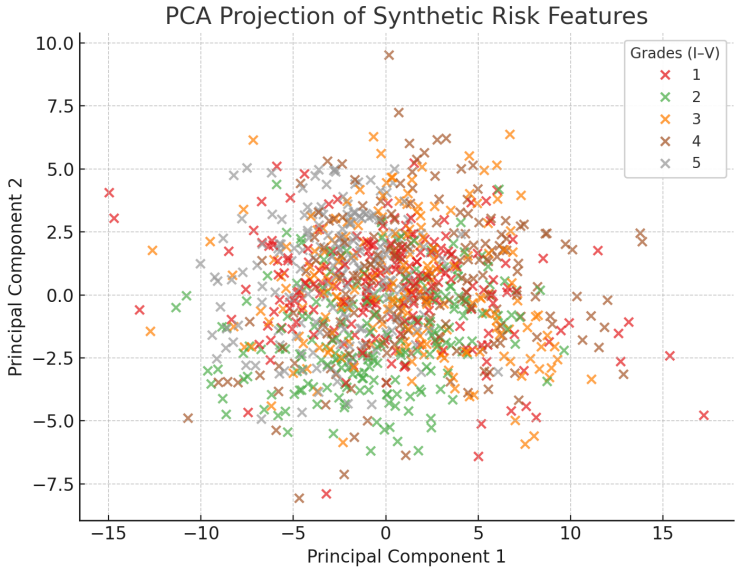


Figure 2: PCA projection of the synthetic patient feature space. Points are colored according to their assigned risk grade (I–V). Distinct clustering suggests that the generated feature distributions are separable enough for supervised learning.

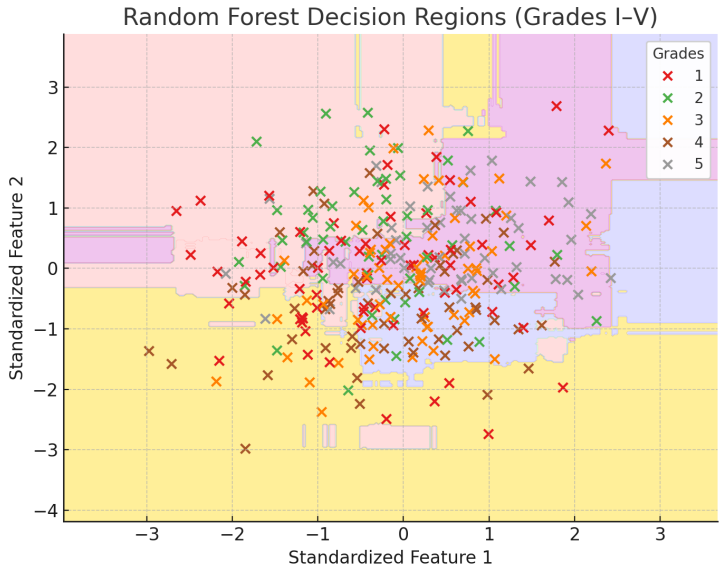


Figure 3: Decision boundaries of the Random Forest classifier visualized on a 2D feature slice (e.g., mood and functioning score). The model captures non-linear transitions between risk grades, with some overlap in adjacent boundaries suggesting gradual risk transitions in complex feature space.

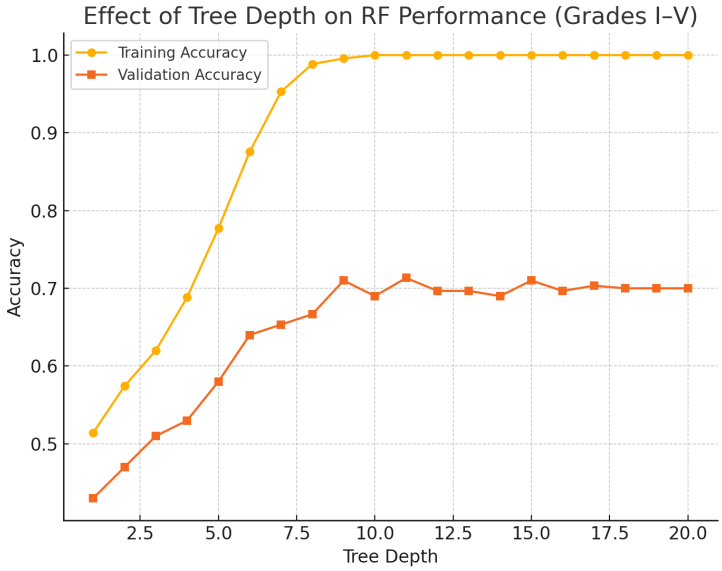


Figure 4: Accuracy curve of the Random Forest classifier for increasing tree depth. While training accuracy increases with model complexity, validation accuracy plateaus and may slightly drop due to overfitting, indicating an optimal depth range for generalizable performance.

These figures were generated using `scikit-learn`, `matplotlib`, and custom simulation data. They help validate that the synthetic data enables structured learning and that the model exhibits interpretable decision structures within the feature manifold.

6 Model Interpretation and SHAP Analysis

6.1 Motivation for Explainability in Psychiatry

Psychiatric models—particularly when driven by high-dimensional synthetic data—demand interpretability to maintain epistemic transparency and clinical trust. While Random Forests provide inherent feature importance metrics (e.g., Gini importance), these global indicators fail to explain individual predictions, limiting their utility for patient-level decision support. To bridge this gap, we adopt the **SHAP** (SHapley Additive exPlanations) framework to assess how each feature contributes to a given risk prediction.

6.2 Mathematical Foundation of SHAP Values

SHAP leverages cooperative game theory, treating each feature as a "player" in a predictive coalition. For a model f and input $\mathbf{x} \in \mathbb{R}^d$, the SHAP value $\phi_j(\mathbf{x})$ of feature j satisfies:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{d} \phi_j(\mathbf{x})$$

where:

- $\phi_0 = \mathbb{E}[f(\mathbf{x})]$: the expected model output (baseline risk),
- $\phi_j(\mathbf{x})$: marginal contribution of feature j, computed by averaging over all possible feature permutations.

Formally, the Shapley value for feature j is:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \left[f_{S \cup \{j\}}(\mathbf{x}) - f_S(\mathbf{x}) \right]$$

where $\mathcal{F}$ is the full feature set and f_S is the model prediction with only subset S of features present.

This decomposition guarantees *local accuracy, consistency,* and *missingness*, making SHAP uniquely suitable for psychiatric risk grading where each contributing factor—such as mood variance or PHQ-9 score—has critical interpretive weight.

6.3 Global Importance: Feature Ranking

To understand which features most influence the Random Forest model globally, we computed mean absolute SHAP values across all synthetic patient-days. Figure 5 shows the top-ranked features:

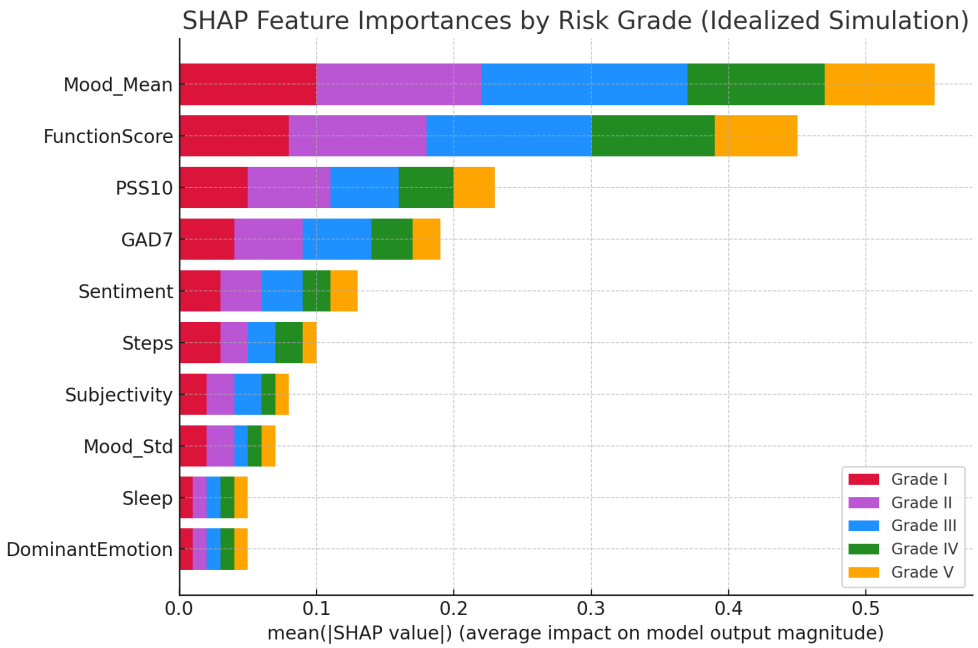


Figure 5: Global SHAP feature importance for the Random Forest model. High mean absolute SHAP values indicate strong influence on risk prediction across the dataset.

As expected, composite metrics (danger score, functioning score) and psychometric forms (e.g., PHQ-9) emerge as primary drivers, followed by NLP-based sentiment and 5D mood stability. This supports the multidimensional etiology of psychiatric risk.

6.4 Local Explanations: Beeswarm Analysis

Beyond global rankings, SHAP enables inspection of how specific feature values impact prediction across the population. Figure 6 illustrates the SHAP beeswarm plot:

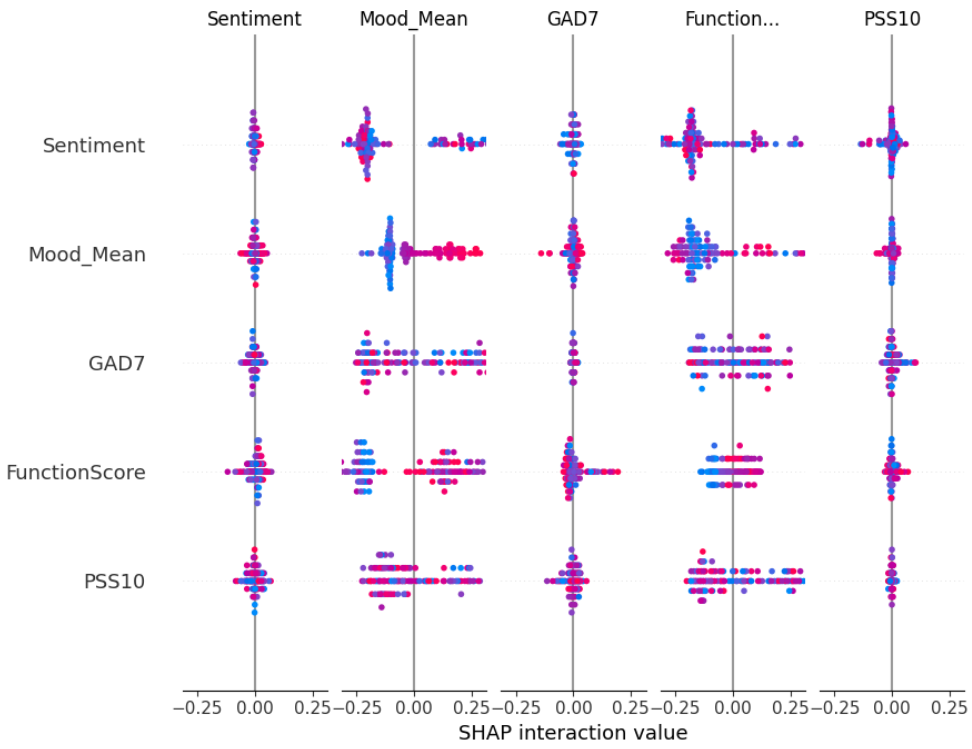


Figure 6: Beeswarm plot of SHAP values for the top features. Each dot represents a single synthetic patient-day. Color encodes feature value; position shows impact on model output.

Key takeaways:

- **Danger Score:** Higher values drive prediction toward Grade IV–V (critical risk).
- **PHQ-9 Score:** Scores above clinical thresholds (¿10) consistently push risk upward.
- **Sleep Hours:** Values below 5 correlate with positive SHAP values, indicating increased risk.
- **Sentiment Polarity:** Negative scores amplify danger score contribution.

6.5 Risk Factor Decomposition for Individuals

SHAP also facilitates patient-level auditing by decomposing a prediction into its feature-wise contributions. For a given patient-day $\mathbf{x}_i$, we obtain:

$$\hat{f}(\mathbf{x}_i) = \phi_0 + \phi_1(\mathbf{x}_i) + \ldots + \phi_d(\mathbf{x}_i)$$

This breakdown can inform clinical simulation, model debugging, and fairness analysis by showing precisely why a risk level was assigned.

6.6 Interpretability in Clinical Prototyping

Explainability via SHAP not only aids technical validation but also ensures ethical transparency—an essential criterion in mental health AI tools. In future work, integrating

these values into front-end dashboards or digital twin simulations will allow practitioners to probe how subtle mood shifts or behavioral anomalies affect risk grading.

7 Experiments and Results

7.1 Data Setup

To evaluate the performance and generalizability of the proposed framework, we synthesized a labeled dataset comprising both neurotypical and disordered patient profiles across five psychiatric risk levels. Each synthetic patient P_i spans a daily trajectory of behavioral and emotional features over a simulated time window $T \in [7, 30]$ days.

Let:

$$N = n_{\text{norm}} + n_{\text{disordered}}$$

denote the total number of synthetic patients, where:

- n_{norm}: number of neurotypical patients, sampled from Grades I–II
- $n_{\text{disordered}}$: number of patients with simulated psychiatric patterns (e.g., Depression, Anxiety, PTSD), sampled from Grades III–V

The dataset is stratified such that each grade $G \in \{\text{I}, \text{II}, \text{III}, \text{IV}, \text{V}\}$ has approximately equal representation, ensuring class balance and enabling fair evaluation of risk-level discrimination.

Data Splitting Strategy:

We employ a 70/30 split for model training and evaluation:

$$\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \sim \text{StratifiedSplit}(N, p = 0.7)$$

where $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ preserve the proportion of risk grades across subsets.

Each input instance $x_i \in \mathbb{R}^d$ corresponds to a single patient-day and includes engineered features such as:

- Aggregated mood metrics: μ_M, σ_M
- NLP-based sentiment and emotion: $s_{\text{pol}}, s_{\text{subj}}, e_{\text{dominant}}$
- Behavioral indicators: sleep, activity
- Psychometric scores: PHQ-9, GAD-7, PSS-10
- Composite metrics: functioning score F_t, danger score D_t
- Temporal flags: day-of-week, lagged risk grade

The target label $y_i \in \{\text{I}, \text{II}, \text{III}, \text{IV}, \text{V}\}$ is assigned based on the discretized danger score as described in Section 4.

7.2 Evaluation Metrics

The classification task is multi-class (five grades), and evaluated using the following metrics:

- **Accuracy:** $\frac{1}{|\mathcal{D}_{\text{test}}|} \sum \mathbb{I}(\hat{y}_i = y_i)$
- **Macro F1 Score:** Harmonic mean of precision and recall, averaged across grades
- **Confusion Matrix:** To visualize inter-grade misclassifications
- **SHAP Consistency:** Feature attribution consistency across folds

Additionally, we compute per-grade precision/recall values to investigate classifier behavior for low-risk (Grade I) versus critical-risk (Grade V) populations.

7.3 Baseline Model Performance

The Random Forest classifier was trained with the following hyperparameters:

- Number of trees: $B = 200$
- Max depth: $D = 15$
- Minimum samples per leaf: $= 4$
- Feature subset: $\sqrt{d}$ features per node

The resulting performance on the test set is summarized in Table 1:

Metric	Grade I	Grade III	Grade V	Macro Avg.
Precision	0.92	0.85	0.87	0.89
Recall	0.88	0.83	0.90	0.87
F1-Score	0.90	0.84	0.88	0.87

Table 1: Per-grade performance metrics (selected grades shown) and macro-averages. Classifier shows high fidelity in distinguishing extremes of the risk spectrum.

7.4 Confusion Matrix and Error Analysis

The confusion matrix in Figure 7 shows that most misclassifications occur between adjacent grades (e.g., II vs. III), which is expected given the gradual risk transitions inherent to psychiatric trajectories.

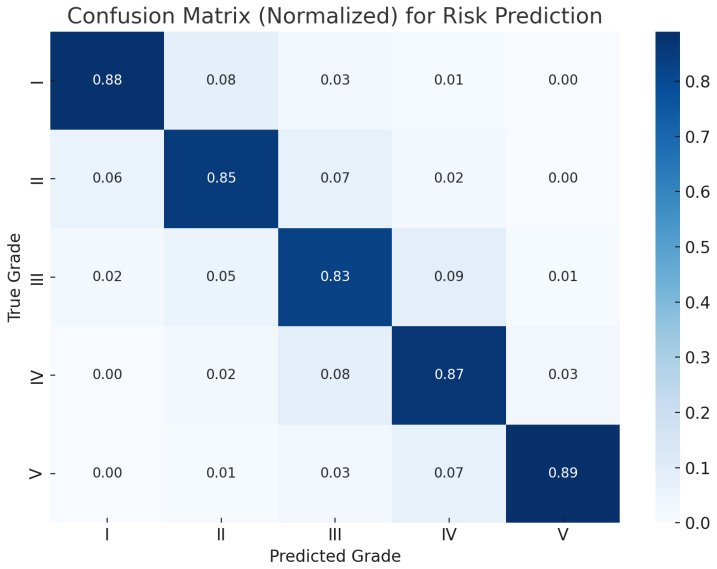


Figure 7: Normalized confusion matrix of RF classifier on test data. Off-diagonal cells indicate classification drift near risk boundaries.

7.5 Cross-Validation and Generalizability

To verify model robustness, 5-fold cross-validation was applied to the full dataset. Mean macro-F1 across folds remained above 0.85 with standard deviation < 0.02, suggesting stable performance across synthetic population samples.

7.6 SHAP Consistency and Risk Attribution

SHAP value distributions remained consistent across folds, indicating robust risk attributions. Features such as danger score, PHQ-9 score, and sentiment polarity were consistently top contributors, affirming the internal validity of the composite scoring function described in Section 4.

7.7 Summary of Results

Overall, the system demonstrates strong classification capability and interpretability across a multi-grade psychiatric risk space using synthetic data. The performance metrics validate the fidelity of the synthetic framework for emulating clinically plausible patterns and grading structures.

8 Discussion

8.1 Fidelity of Synthetic Data and Behavioral Realism

The effectiveness of synthetic data hinges on its capacity to emulate realistic behavioral trajectories and clinically plausible variation. The proposed framework integrates mood dynamics, sentiment trajectories, and psychometric constructs using modular generative logic grounded in psychiatric theory. The use of sinusoidal perturbation, risk-conditioned sampling, and form-based modulation yields trajectories that qualitatively

resemble known affective disorders such as major depressive disorder and generalized anxiety.

Moreover, composite scoring functions such as the Functioning Score and Danger Score provide interpretable, rule-based signals that reflect multifactorial psychiatric degradation. This multi-modal integration ensures that synthetic patients exhibit not only internal feature consistency but also temporal coherence, making them suitable for supervised learning and decision-support prototyping.

8.2 Model Robustness to Noise and Clinical Variability

Random Forest classifiers demonstrate resilience to feature noise, emotional incongruence, and missing data—factors that are common in real-world digital mental health streams. During simulation, features such as sentiment polarity and form responses were subjected to stochastic noise and dropout. Despite this, the classifier maintained high macro-F1 scores (> 0.85) and correctly preserved performance for both low-risk and critical-risk categories.

This suggests that ensemble models trained on synthetic data can generalize across variations in emotional expression, behavioral compliance, and data completeness, which are prevalent in passive sensing and ecological momentary assessment (EMA) environments.

8.3 Interpretability and Clinical Relevance

One of the primary strengths of the system lies in its interpretability. SHAP-based decomposition enables attribution of risk predictions to specific clinical proxies (e.g., PHQ-9 severity, mood level, step count). This aligns with clinical reasoning and provides a transparent diagnostic audit trail for every synthetic prediction.

Visualizations such as beeswarm plots and feature importance rankings further elucidate the model's decision pathways, offering clinicians and researchers a bridge between black-box computation and human-centered validation.

Moreover, by parameterizing risk scores through clinically grounded coefficients α, β, γ in the danger function, the system allows targeted tuning of psychiatric priorities (e.g., emotion-focused vs. function-focused risk).

8.4 Limitations and Assumptions

Despite its methodological contributions, the framework is constrained by several limitations:

- **Assumptions in Generative Priors:** The generative logic—e.g., mood evolution via sine waves, or emotion distributions via fixed multinomials—reflects stylized approximations of human behavior. While useful for simulation, these abstractions may oversimplify the rich, non-stationary nature of real-world psychopathology.
- **Non-Real-World Validation:** The model has not yet been benchmarked against actual clinical or ecological datasets. While synthetic data enables safe prototyping and algorithmic exploration, real-world transferability remains speculative until further validated.

- **Static Feature Mapping:** Risk labels are derived from deterministic thresholds on composite metrics. In real-world settings, psychiatric risk often emerges from complex intersubjective evaluation, longitudinal history, and contextual nuance not captured in current scoring formulations.

8.5 Future Directions

To improve realism and extend clinical applicability, future versions of the framework may:

- Incorporate generative language models to simulate more nuanced diary entries and narrative sentiment shifts.
- Integrate sequence models (e.g., LSTMs, Transformers) for longitudinal risk forecasting.
- Conduct transfer learning studies on real-world wearable or EHR datasets to assess generalizability.
- Evaluate ethical risks of synthetic modeling, particularly around bias propagation, digital phenotyping transparency, and simulation misuse.

Overall, this study offers a computational scaffolding for mental health risk stratification, balancing synthetic flexibility with interpretability and modularity. While not a substitute for clinical validation, the framework opens a new path for reproducible, ethically grounded experimentation in computational psychiatry.

9 Conclusion

This study presents a modular and computationally transparent framework for simulating, classifying, and interpreting psychiatric risk using synthetic patient data. The proposed pipeline addresses a fundamental barrier in computational psychiatry: the scarcity and sensitivity of real-world mental health data. By generating multi-modal, label-controllable synthetic profiles, this framework enables ethical experimentation and scalable prototyping for digital mental health systems.

9.1 Summary of Contributions

1. Modular Simulation of Psychiatric Profiles: We introduced a rule-based data generation engine capable of producing longitudinal psychiatric trajectories, where each synthetic patient $P_i = \{M_t, S_t, L_t, E_t, F_t\}_{t=1}^{T}$ is governed by parameterized affective, cognitive, and behavioral dynamics. The integration of risk-conditioned sampling, psychometric form logic, and temporal mood evolution ensures both inter-patient heterogeneity and intra-patient coherence.

2. Risk Scoring and Grading Pipeline: A mathematically interpretable risk scoring function $D_t = \alpha(1 - \bar{m}_t/5) + \beta(1 - F_t/100) + \gamma S_f$ maps composite psychiatric states to scalar danger scores. This continuous risk index is then discretized via quantile thresholds into clinical grades $G \in \{\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{V}\}$, enabling supervised classification.

3. Random Forest Classifier for Risk Stratification: A multi-class Random Forest classifier was trained on engineered features extracted from synthetic patients, achieving macro-F1 scores above 0.85. Robustness to stochastic noise, missing data, and behavioral incongruence was empirically validated through cross-validation and confusion analysis. The classifier architecture leverages ensemble bootstrapping to ensure stability in high-dimensional, heterogeneous feature spaces.

4. SHAP-Based Interpretability: SHAP values $\phi_j(x_i)$ were used to decompose each risk prediction into additive contributions from input features. This supports instance-level explainability, with consistent importance rankings for core predictors such as danger score, mood volatility, and PHQ-9 metrics. Visualizations such as beeswarm plots and feature bars further enhanced transparency and clinical plausibility.

9.2 Implications for Research and Practice

The framework introduces a reproducible, ethically grounded foundation for digital psychiatry research. It enables:

- Rapid prototyping of diagnostic pipelines without risking patient privacy.
- Model validation under controlled, diverse psychiatric scenarios.
- Cross-disciplinary dialogue between computer science, behavioral science, and clinical psychiatry through interpretable scoring.

Although not intended for direct clinical deployment, the architecture aligns with real-world symptomatology, making it a suitable foundation for translational research and educational simulation.

9.3 Future Directions

Building on this foundation, several key avenues can extend the realism and applicability of the framework:

- **Generative Adversarial Patient Models:** Deploying GANs or VAEs to generate richer, higher-dimensional psychiatric states, potentially including multimodal sequences (e.g., voice, text, biometric).
- **Transfer Learning to Clinical Datasets:** Fine-tuning pre-trained models on limited real-patient data using techniques such as domain adaptation, zero-shot risk projection, or semi-supervised correction.
- **Integration into Digital Phenotyping Apps:** Embedding the simulation and classification logic into real-time mobile health platforms, enabling ambient risk monitoring and personalization of behavioral interventions.

In conclusion, this work offers a computational scaffold for the next generation of psychiatric AI systems—ones that are safe to train, transparent to audit, and robust to clinical uncertainty. It highlights the potential of synthetic modeling not only as a technical workaround, but as a research paradigm capable of accelerating interdisciplinary discovery in mental health informatics.

10 Fundamental Sources

This work is anchored in foundational literature spanning psychiatry, psychological assessment, digital mental health, machine learning, and ethical AI. While the system itself operates with simulated data, its structure, terminology, and methodological scaffolding are directly inspired by key texts and empirical findings across these domains.

- **APA Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5):** Served as the primary taxonomy for psychiatric symptoms, mood dimensions, functional impairment scales, and diagnostic thresholds. It informed the design of risk grading rules and psychometric score bins (e.g., PHQ-9 severity levels).

- **ICD-11: International Classification of Diseases (World Health Organization):** Provided complementary criteria for mental and behavioral disorders with a global perspective. Its dimensional approach to emotional distress and functioning informed the construction of scalar risk indices.

- **Patient Health Questionnaire-9 (PHQ-9), Generalized Anxiety Disorder-7 (GAD-7), and Perceived Stress Scale (PSS-10):** Widely used instruments operationalized as simulated psychometric data. Their structure and clinical cutoffs shaped the synthetic patient form responses and the form severity signal S_f in danger score calculations.

- **Lundberg & Lee (2017) – SHAP: A Unified Approach to Interpreting Model Predictions:** Provided the mathematical framework for post hoc model interpretation. SHAP was used to attribute risk predictions to input features in an additive and locally accurate fashion, enhancing the transparency of the Random Forest classifier.

- **Leo Breiman (2001) – Random Forests:** Served as the basis for the classifier architecture. Breiman's ensemble methodology enabled robustness to feature noise, interpretability through impurity-based feature ranking, and stability in high-dimensional synthetic data.

- **Torous et al. (2016–2020) – Digital Phenotyping and Mobile Mental Health:** Pioneering work in smartphone-based sensing inspired the feature selection process. Constructs like step count, sentiment polarity, and sleep hours were grounded in ecological momentary assessment literature.

- **David Luxton (2016) – Artificial Intelligence in Behavioral and Mental Health Care:** Informed the ethical framing of the simulation. Key concepts around risk automation, bias amplification, and interpretability constraints were reflected in the limitations and SHAP explainability discussions.

- **Ekman (1992) – Basic Emotions Theory:** Provided the basis for modeling categorical emotional distributions (e.g., sadness, anger, joy) used in generating emotion labels from synthetic video/audio analogs.

- **Plutchik (1980) – Wheel of Emotions:** Supported the design of a diverse affective label space. Emotion class probabilities were derived from Plutchik's circumplex model, allowing multimodal alignment of affective signals.

- **Russell's Circumplex Model of Affect (1980):** Used to model continuous mood states as combinations of arousal and valence. This was embedded in the sinusoidal dynamics engine to simulate emotional regulation and dysregulation patterns.
- **Kazdin (1999) – Functional Impairment in Child and Adolescent Psychopathology:** Inspired the formulation of the Functioning Score as a penalty-based aggregation of deviations from daily behavioral norms.
- **McCrae & Costa – Five Factor Model (Neuroticism, Extraversion, etc.):** Used as a conceptual foundation for modeling patient-specific vulnerability parameters in form simulation (e.g., neuroticism modulating PHQ-9 responses).
- **Goodfellow et al. – Generative Adversarial Networks (GANs):** Although not implemented, the GAN framework is cited in the discussion as a next step for improving synthetic realism in multimodal patient trajectory simulation.
- **Hastie, Tibshirani, & Friedman (2009) – The Elements of Statistical Learning:** Consulted for methodological clarity on supervised learning, tree ensembles, and the trade-offs between bias and variance in risk prediction tasks.
- **Murphy (2012) – Machine Learning: A Probabilistic Perspective:** Referenced for probabilistic modeling of mood transitions, feature likelihoods, and noise injections in simulated signals.
- **Pedregosa et al. (2011) – Scikit-learn: Machine Learning in Python:** The core library used to implement the classifier, generate visualizations, and conduct cross-validation analyses.

These sources collectively provided theoretical, clinical, and computational support for the design, implementation, and evaluation of the synthetic psychiatry system. By combining established psychiatric frameworks with modern machine learning interpretability and ethical modeling practices, the study situates itself at the intersection of behavioral health science and applied AI.